

## Offline, once: metric mapping + persistent instance registration

Multiview RGB → COLMAP + metric frame → Metric 3DGS training → SAM + cross-view association → Persistent IDs + one-time naming

### Wearer gaze → world fixations

 **Localize camera**  
Surveyed ArUco

 **Lift gaze to 3D**  
Pose + depth

 **Form fixations**  
Space + time

  
World-frame events

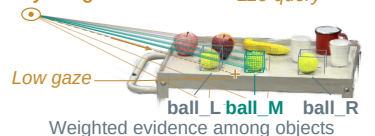
### 3DGS-based instance disambiguation

A low gaze still supports the target

**Eye origin**

$\pm 2\sigma$  query

*Low gaze*



ball\_M 67% ball\_L 16% others 17%

**Visible mass**

$q_k = m_k / W$

**Normalize**

$S_k = q_k / C_k$

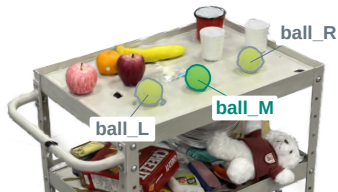
**Rank + gate**

ID / reject

Size + distance · candidate competition

### Shared 3DGS instance map

One metric frame for user and robot



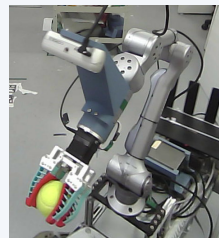
**Depth + opacity + instance labels**

Persistent IDs + metric locations

### Robot execution

Grounded target → action

**Navigate in map frame**  
Reobserve target



**Grasp + carry**

**Ordered commands**  
move · pick · place

### Speech → grounded commands

**Streaming ASR**

Word timestamps

**LLM: actions + slots**

"Bring me this ball."

**Word-fixation binding**

gaze  
word

ball\_M

"this"

**Compile + dispatch**

Action + ID + world arguments  
**On successful grounding**